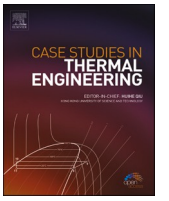
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# Experimental investigations of heat transfer, energy, and exergy-based sustainability of a novel photovoltaic thermal system

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ABSTRACT

The inefficiency of photovoltaic systems is a major obstacle. This research proposes an advanced collector design with dimpled and petal-patterned absorber tubes, coiled twisted tape, and nanofluids combined with nanophase changing materials. The adopted methodology consists of two phases, namely experimentally characterizing the heat transfer performance of the absorber tubes and conducting indoor experiments to evaluate the performance of the new photovoltaic thermal system. The experiments were performed under different flow rates of (0.01–0.085 kg/s), irradiances (400–1000 W/m$^2$), and six different coolants. The initial experiment revealed an inverse relationship between the mass flow rate and the thermal performance of the absorber tubes. However, mass flow rates, solar irradiances up to 1000 W/m$^2$, and using various coolants positively impacted the overall performance of the photovoltaic system. The absorber tube with dimples, petal arrays, coiled twisted tape, and nanofluid outperformed the smooth tube with water threefold. Additionally, the photovoltaic thermal system utilizing nanofluids and nanophase changing materials achieved electrical and thermal energy enhancements of 32 % and 21.2 %. The optimal design demonstrated environmental and economic viability, with total output surpassing input energy by 2.11 MWh and net $CO_2$ mitigation exceeding $CO_2$ emissions by 0.63 tons.

## 1. Introduction

Global warming, environmental pollution, greenhouse gas emissions, and the depletion of conventional energy sources are among the most pressing challenges of the modern era [1]. These issues have prompted collaborative efforts among governments, international organizations, and scientific communities to seek sustainable solutions [2]. Solar technology is widely recognized as one of the most sustainable energy production methods, valued for its effectiveness in electricity generation, cost efficiency, and environmentally friendly characteristics [3]. Over the past five decades, significant attention has been directed toward solar technology as a viable alternative to traditional energy sources, particularly fossil fuels [4]. Nevertheless, numerous studies have reported a decline in the electrical efficiency of photovoltaic (PV) systems due to heat accumulation on their surfaces. This phenomenon has been quantified, with efficiency decreasing by approximately 5 % for every 10 °C increase in surface temperature, primarily attributed to the

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temperature coefficient of the surface material [5]. The crucial impact of this issue has driven the development of Photovoltaic Thermal (PVT) systems, which integrate thermal collectors to the back of PV modules to substantially reduce surface temperature, enhance electrical efficiency, and extend the overall lifespan [6]. In addition to improving electrical efficiency, PVT systems have demonstrated greater cost-effectiveness, reduced spatial footprint, and the added benefit of producing hot water [7]. With the numerous advantages of PVT systems, the field has witnessed significant advancements, particularly by integrating passive and active cooling techniques [8]. Active cooling methods rely on external energy sources or mechanical systems, whereas passive cooling techniques, increasingly favored in the current literature, utilize natural principles to enhance heat transfer. These methods include enhancing the surface area, generating turbulence, forming a vortex, and improving material properties at the microstructural level [9]. Commonly employed thermal management strategies in recent studies include nanofluids (NF), nanophase changing materials (NPCM), dimples, petal arrays, coils, and twisted tapes.

Integrating NFs into PVT systems has become a highly effective approach to lowering surface temperature [10]. Due to water's inherently low thermal conductivity, nanoparticles such as metals and metal oxides are mixed with base fluids like water to enhance their thermophysical properties significantly. For instance, M. Sheikholeslami and Z. Khalili [11] have utilized non-toxic graphene nanoplatelets-based NF in weight concentrations of 0.025 % and 0.1 %. They have tested their PVT system under primary conditions of flow rates of 0.01–0.18 m/s and solar irradiance of 800 W/m$^2$, and they have reported optimal exergy, energy, and thermal efficiencies of 15.32 %, 14.05 %, and 55.22 %, respectively. In another similar study, M. Alktranee et al. [12] compared the effects of $TiO_2$-enhanced NF, $Fe_2O_3$- enhanced NF, and a hybrid $TiO_2$- $Fe_2O_3$-enhanced NF, and they found that the hybrid NF with a concentration of 0.3 % caused enhancement of 13 % in electrical efficiency and 44 % in thermal efficiency. Moreover, B. Srimanickam et al. [13] employed cerium oxide-enhanced NF in a PVT system, and they reported that the new NF can improve overall efficiency between 5 and 10 % compared to the cases using water alone. Additionally, S. Sreekumar et al. [14] have compared the effects of Carbon quantum dot (C-dot), MXene, and MXene/C-dot hybrid NFs from economic and environmental viewpoints. Their results demonstrated that the three NFs caused $CO_2$ emissions of 2.024, 1.979, and 1.931 kg CO2/day, respectively, while the hybrid one reduced emission cost by $ 0.02/day. Finally, F.A. Radhi et al. [46] conducted an experiment on a newly designed collector using 0.2 % $Al_2O_3$. Their results indicate that in the best-case scenario, there was an electrical efficiency enhancement of 39.24 % compared to the PV alone. In conclusion, despite minor disadvantages such as higher initial costs and potential environmental impacts, the widespread adoption of NFs in PVT applications is inevitable due to their significant benefits [15].

Phase-Changing Materials (PCMs) are widely integrated into PVT systems due to their remarkable ability to absorb and release substantial amounts of latent heat during phase transitions [16]. However, their inherently low thermal conductivity has driven the development of nanophase changing materials (NPCMs), which incorporate nanoparticles, such as metals and metal oxides, into their microstructures to enhance thermal performance [17]. For example, M.M. Maseer et al. [18] have integrated $Al_2O_3$-enhanced PCM into their newly designed PVT system under mass flow rates of 1–5 LPM. They reported that their new design outperforms a standalone PV by 14.04 %. Similarly, S. Al Arni et al. [19] compared the performance of three PV modules cooled by one, two, and three layers of graphene-enhanced PCM. The three-layer degraded configuration consistently exceeded the performance of other setups, keeping PV temperatures up to 5 °C lower and electrical efficiencies up to 4 % higher during 100 min of operation. Bhutto, Y. A. et al. [20] studied the effect of organic PCM at a flow rate and an irradiance of 0.45 MPL and 440 W/m$^2$. The proposed design significantly reduced surface temperature from 59 °C to 36 °C, demonstrating the PCM's effectiveness in thermal regulation under these conditions. Finally, Md.G. Kibria et al. [21] investigated the impact of $Al_2O_2$ and ZnO-enhanced PCM in photovoltaic thermal (PVT) systems. The findings confirm that the system achieved an optimal thermal efficiency of 51.28 %, accompanied by a notable enhancement in electrical efficiency, recorded at 31.46 %.

While incorporating NFs and NPCMs is highly effective in reducing surface temperature in PVT systems, these systems typically depend on external energy sources for operation. However, recent literature has increasingly emphasized passive cooling techniques to enhance the overall performance of PVT systems. Approaches such as utilizing roughened surfaces and integrating insertion devices have significantly improved performance without relying on external energy inputs [22]. S.A.A. Mehrjardi et al. [23] examined the performance of a tube featuring tear-drop dimples, and they concluded that their new design increased the performance of the absorber tube by 51.5 compared to a smooth tube. In another study, Kaood, A [24]. examined the impact of various dimple shapes on heat transfer enhancement, such as spherical, cylindrical, conical, and step-conical. The results indicate that the step-conical dimple shape, at a Reynolds number of 3000, achieves the optimal thermohydraulic performance factor (THPF), with a 7.81 % improvement in heat transfer efficiency. Additionally, A. Khashaei et al. [25] examined the performance of a dimpled tube channeling $Al_2O_3$ NF under Reynolds numbers 500 to 2250 and a constant heat flux of 10,000 W/m$^2$. They reported that their new design recorded enhancement by 3.42 times in heat transfer coefficient. Munusamy. A et al. [47] designed a new collector of a parabolic plate solar water heater featuring dimples on its surface and aluminium coating. Their findings show that thermal efficiency can be enhanced by 31.85 % at optimal testing conditions. Finally, M. Arun et al. [26] studied the effects of $TiO_2$-enhanced NF and a dimpled tube on the performance of a trough solar collector under mass flow rates of 0.5 kg/min to 2.5 kg/min. The results indicated that the system's overall performance was increased by 34.25 % under the effect of the cooling techniques. Therefore, it can be concluded that incorporating dimples into the design of absorber tubes enhances heat transfer by inducing turbulence and increasing the contact surface area.

Incorporating insert devices, such as coils or twisted tapes, is another effective technique for improving the thermal performance of a tube. While increasing the contact area, these devices also promote the formation of vortexes within the flow, significantly improving heat transfer [27]. W. Yang et al. [28] investigated the influence of twist ratio on thermometric generators and found that a twist ratio ranging from π to 3π doubled THPF under optimal testing conditions. Similarly, Vishwakarma D.K. et al. [29] analyzed the impact of spring ratio in a circular tube, considering Reynolds numbers from 502 to 10,936 and heat fluxes of 2, 3, and 4 kW/m$^2$. Their results

revealed that a spring ratio of 3 causes a transition within Reynolds numbers between 661 and 1734. In contrast, a spring ratio of 5 induces a transition within the Reynolds number range of 1305 to 2226. In another study, H. Luqman Azeez et al. [30] investigated the performance of a PVT system that incorporated a coiled twisted tape, and they found that the optimal overall energy and exergy results were recorded at 98.4 % and 15.7 %, respectively. Alsharifi.T et al. [48] designed a new PVT collector incorporating a porous twisted tape. The study demonstrates that the proposed design increases the electrical and thermal efficiency of the PVT by 2.27 % and 14.85, respectively, at an optimal testing condition. Finally, B.O. Ahmed et al. [31] also adopted a coiled twisted tape, and they also reported that the system increased thermal efficiency by 25 % due to the presence of the newly designed inserter. Therefore, integrating insert devices into the design of absorber tubes can substantially enhance their performance.

Recent literature highlights the synergistic effect of integrating multiple cooling techniques into PVT systems [32]. For example, A. B. Al-Aasam et al. [33] combined a twisted tube, silicon carbide-enhanced NF, and silicon carbide-enhanced PCM in a novel PVT design. Their system achieved electrical, thermal, and overall efficiencies of 9.57 %, 84.74 %, and 94.31 %, respectively. Similarly, A. M. Bassam et al. [34] demonstrated that a PVT system could be enhanced by incorporating cooling techniques such as micro-fin, twisted tape, and silicon carbide-enhanced NF and PCM. This combination resulted in maximum electrical and thermal efficiencies of 10.8 % and 83.3 %, respectively. These studies underscore the potential of integrating various passive cooling strategies, which, when effectively optimized, can lead to significant improvements in system performance.

Despite significant advancements in the field of PVT systems, several research gaps remain to be addressed. For instance, while the designs by A.B. Al-Aasam et al. and A.M. Bassam et al. [33,34] are among the most recent and effective, they still require further optimization to enhance power output. Currently, the power production of these systems is limited to 20 W for a panel with an output power rating of 30 W. The novelty of this study lies in the design of a solar collector featuring surface dimples and a petal pattern, integrated with spiral twisted tape, utilizing SiC-enhanced NF, and encapsulated with SiC-enhanced PCM. This design is anticipated to outperform previous designs, particularly the benchmark cases referenced in Refs. [33,34], and align with the key sustainability principles, such as cost-effectiveness and environmental impact. The study will first characterize the heat transfer performance of the newly developed absorber tubes. Subsequently, a PVT system will be fabricated using this absorber tube, and the performance and sustainability feasibility of the new PVT system will be meticulously evaluated. This innovative design is expected to facilitate more effective integration of photovoltaic technology into the energy infrastructure of countries with high solar irradiance, alongside other applications such as hot water production.

## 2. Methodology

The study will be divided into two experimental phases.

1. **The first phase:** heat transfer characterization of the following absorber tubes, will be revealed.
   - Smooth tube (ST)
   - Dimpled and petaled array absorber tube (DPAT)
   - Dimpled and petaled array absorber tube with a coil (DPATC)
   - Dimpled and petaled array absorber tube with a twisted tape (DPATTT)
   - Dimpled and petal arrays absorber tube with a coiled twisted tape (DPATCTT)
2. **The second phase:** the best absorber tube will be used to engineer a collector for a PVT system, and its effectiveness will be tested with the following coolants
   - Water at flow rates of (0.01–0.085) kg/s and solar irradiance (400–1000) W/m$^2$
   - 0.3 % VC of SiC-enhanced NF at flow rates of (0.01–0.085) kg/s and solar irradiance (400–1000) W/m$^2$
   - 0.6 % VC of SiC-enhanced NF at flow rates of (0.01–0.085) kg/s and solar irradiance (400–1000) W/m$^2$
   - NPCM with Water at flow rates of (0.01–0.085) kg/s and solar irradiance (400–1000) W/m$^2$
   - NPCM with 0.3 % VC of SiC-enhanced NF at flow rates of (0.01–0.085) kg/s and solar irradiance (400–1000) W/m$^2$
   - NPCM with 0.6 % VC of SiC-enhanced NF at flow rates of (0.01–0.085) kg/s and solar irradiance (400–1000) W/m$^2$

As indicated above, mass flow rates ranging (from 0.01 to 0.085) kg/s and solar irradiance (400–1000) W/m$^2$ were selected as lower and upper limits of the study, reflecting the real applications of PVT, as supported by references number [31,33,and34]]. Moreover, the choice of NF and NPCM with their respective concentrations are also taken from the same references, considering their high efficiency, cost-effectiveness, and sustainability.

### *2.1. The proposed design of the photovoltaic thermal PVT system*

The newly designed photovoltaic thermal (PVT) system in this study consists of four main components: a photovoltaic module, a novel thermal collector incorporating dimples, petal arrays, and a spiral twisted tape, an advanced heat transfer fluid, and a thermal storage unit utilizing PCM.

#### *2.1.1. The photovoltaic module*

In this study, the photovoltaic module employed is the SW30M − 36, a monocrystalline photovoltaic panel. The comprehensive electrical and physical specifications of the SW30M − 36 are provided in Table 1.

*2.1.2. The thermal collector*

The tube design shown in Fig. 1 features dimples and petal patterns on its surfaces. Such features facilitate the creation of turbulence and increase the contact area, contributing significantly to enhanced heat transfer. Moreover, it is equipped with a spiral twisted tape, increasing heat transfer through the contact area and inducing vortex generation. Nine pipes of 30 cm are configured in a serpentine shape at the rear of the PV to reduce the surface temperature [33]. The choice of number of pipes and configuration is taken from the benchmark studies, which have used the same number of pipes and configuration. Moreover, other conditions including the choices of mass flow rates, irradiances, concentrations of nanoparticles in NF, and concentration of nanoparticles in PCM, are also taken from the same studies.

*2.1.3. The preparation, thermophysical properties, and characterization of the nano-solution*

The study used the silicon-carbide nanoparticles detailed in Table 2 to prepare two concentrations of NFs. In the preparation of 0.6 % VC of SiC-enhanced NF, 463.2 g of the nanoparticles were mixed with every 24 L of water, while 232.6 g was employed to prepare 0.3 % VC of SiC-enhanced NF [30]. Since the solubility of the nanoparticle is weak, polyvinylpyrrolidone was employed as a surfactant. The mixture was mechanically agitated for 10 min before undergoing ultrasonic treatment in an 80 °C bath with a 50 Hz vibration frequency for 3 h to achieve uniformity. Fig. 2 (A) is the preparation procedure of the NF [30].

The main issues in NF preparations are stability issues, poor mix, low thermal conductivity, and lack of homogeneity. This study conducted Field Emission Scanning Electron Microscope (FESEM) and Transmission Electron Microscopy (TEM) analysis of the samples, as shown in Fig. 2 (B), aside from recording their thermophysical properties after one month from their preparation to ensure their correct preparation and effectiveness. The FESEM and TEM images reveal that the mixture has good stability with a concentration of 50.3 % for Si, 42.2 % for C, and 6.3 % for Ox. The FESEM and TEM results of this study match those of studies in Refs. [35,36]. Additionally, as shown in Fig. 2 (C), The Anton Paar model DMA 1001 Density Tester, Anton Paar model MCR 92, and Linseis model THB-500 were used to calculate thermophysical properties such as density, viscosity, and thermal conductivity. The results were confirmed numerically through Eqs (2)–(6) [30and31]. The results show significant improvements in all the properties, guaranteeing enhanced thermal efficiency. The FESEM, TEM, thermophysical properties, and lack of sedimentation after one month are good proof that the NFs are prepared correctly and will be effective during the experiment.

$$\phi = \frac{\frac{m_p}{\rho_{np}}}{\frac{m_p}{\rho_{np}} + \frac{m_f}{\rho_f}} \tag{1}$$

Eq. (1) was used to determine the volume concentration of nanoparticles that were used in the nano-solution where $\phi$ is the volume concentration of nanoparticles, $m_p$ and $m_f$ are the mass of the nanoparticle and the base fluid, and $\rho_f$ *and* $\rho_{np}$ are the density of the nanoparticle and the base fluid.

$$\rho_{nf} = (1-\phi)\rho_{bf} + \phi\rho_{np} \tag{2}$$

Eq. (2) is used to calculate the density of the nanofluid where all the terms are defined previously.

$$\mu_{nf} = \mu_{bf} \times \frac{1}{\left(1 - 3.87\left(\frac{\rho_{np}}{\rho_{bf}}\right)^{-0.3} \times \phi^{1.03}\right.} \tag{3}$$

$$\rho_{nf} = \frac{6M}{N \times \pi \times \rho_{nb}} \tag{4}$$

Eqs. (3) and (4) were used to find the viscosity of the nanofluid, where $\mu_{nf}$ *and* $\mu_{bf}$ are the viscosity of the nanofluid and the base fluid, $\rho_{nf}, \rho_{nb}$, *and* $\rho_{np}$ are the densities of the nanoparticle, nanofluid, and based fluid, N Avogadro number, and M mass of the nanoparticle.

$$K_{nf} = \frac{K_{np} + 2K_{bf} + 2\phi(K_{np} - K_{bf})}{K_{np} + 2K_{bf} - 2\phi(K_{np} - K_{bf})} \times K_{bf} \tag{5}$$

Eq. (5) was used to determine the thermal conductivity of the nanofluid, where k represents the thermal conductivity.

**Table 1**
The physical and electrical specifications of the used PV module.

| Photovoltaic specification | |
|---|---|
| Module brand name | SW30M-36 |
| Type of PV cell | Mono |
| Maximum power, maximum voltage, maximum current | 30 W, 18 V, and 1.66 A |
| The voltage of the open circuit and current of the short circuit | 22.9 V and 1.99 V |
| Size of the panel | 360*670*17 mm |
| Standard Test Condition | 1000 W/m$^2$, 1.5AM, 25 °C |
| Output tolerance | 5(W) |

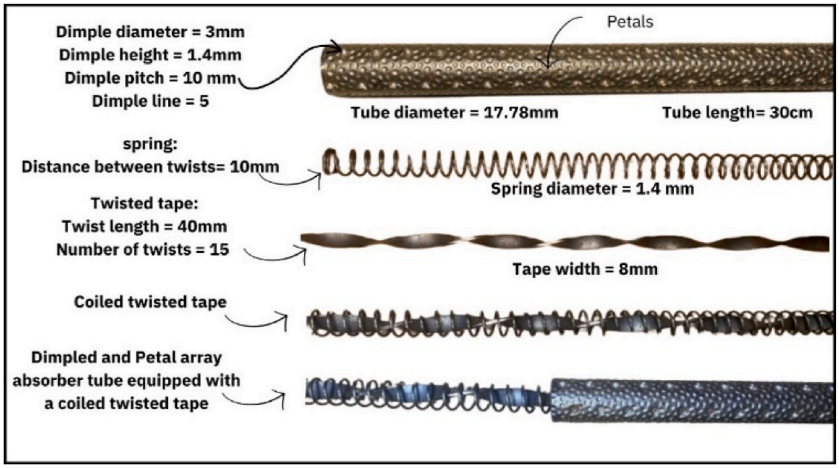



**Fig. 1.** The geometrical details and pictures of the used tube.

**Table 2**
Specifications of the nanoparticles utilized in the experiment.

| Nanoparticle details | |
|---|---|
| Production | USA. Nanoparticle |
| Color | Greyish to black |
| Quality | 99+% |
| PH level | 3-7 at 20 °C |
| Crystal shape | Cubic |
| Melting degree | 2745 °C |
| particle scale | 41–61 nm |
| Heat Conductivity | 371–491 W/m.K |
| Apparent Density | 3.21 g/cm$^3$ |
| Electronic potential | 27.9 mV |

$$CP_{mix} = \frac{(1-\phi)(\rho CP)_{water} + (\phi \rho CP)_{np}}{\rho_{mix}} \quad (6)$$

Finally, Eq. (6) was used to calculate the specific heat capacity of the nanofluid, which is represented by CP.

The same nanoparticle used in the preparation of the NFs was used in the fabrication of the NPCM. Paraffin wax detailed in Table 3 was melted in an ultrasonic device and mixed with the nanoparticle. 100 g of nanoparticles were used with every 2.5 kg of PCM [31]. Mechanical agitation was performed for 10 min, followed by ultrasonic treatment in an 80 °C bath with a 50 Hz vibration frequency for 3 h [31]. Fig. 3 shows the preparation steps and FESEM images of the NPCM, which show the homogeneity and correct preparation of the samples.

It should be noted that the prepared nanofluids are expected to significantly enhance the PVT system's overall performance. As shown in Fig. 2 (C), the thermophysical properties of the nanofluid exhibit a positive trend within the experimental temperature range. Furthermore, the PCM has a melting temperature range of 45–59 °C, which is well-suited for the temperature boundaries of this study, as it will be discussed in later sections.

### 2.2. The experimental rigs

Two experiments were conducted to first release the heat transfer characterization of the novel absorber tubes and second test its effectiveness as a collector in a PVT system. Five thermocouples were mounted on the pipe wall to measure temperatures at different points, while a silicon rubber heater (400 W/m$^2$) and insulation provided heat flux. Two RTD sensors at the inlet and outlet monitored fluid temperatures, all linked to an AT47V data logger for temperature tracking. A DPG 409 gauge measured pressure drops across the sample. Fluid circulation was managed using a cooler, mass flow meter, heat exchanger, and pumps, with heat flux controlled via a voltage regulator. Inputs included mass flow rate and heat flux, while outputs measured wall temperatures, fluid temperatures, and pressure drops. Fig. 4 (A and B) illustrates the setup.

The second experiment used an indoor solar simulator, as shown in Fig. 4(C and D), with key instruments including a TES 132 pyranometer, DHYB-800 flow meter, DT80 data logger, MP-11 I-V checker, cooling unit, RTD sensors, K-type thermocouples, motors, and fluid tanks for water or NF. The fluid was stored at ambient temperature using a chiller and heat exchanger, then circulated by a motor, with flow rates (0.01–0.085 kg/s) measured by a flow meter. RTD sensors monitored inlet and outlet fluid temperatures, while

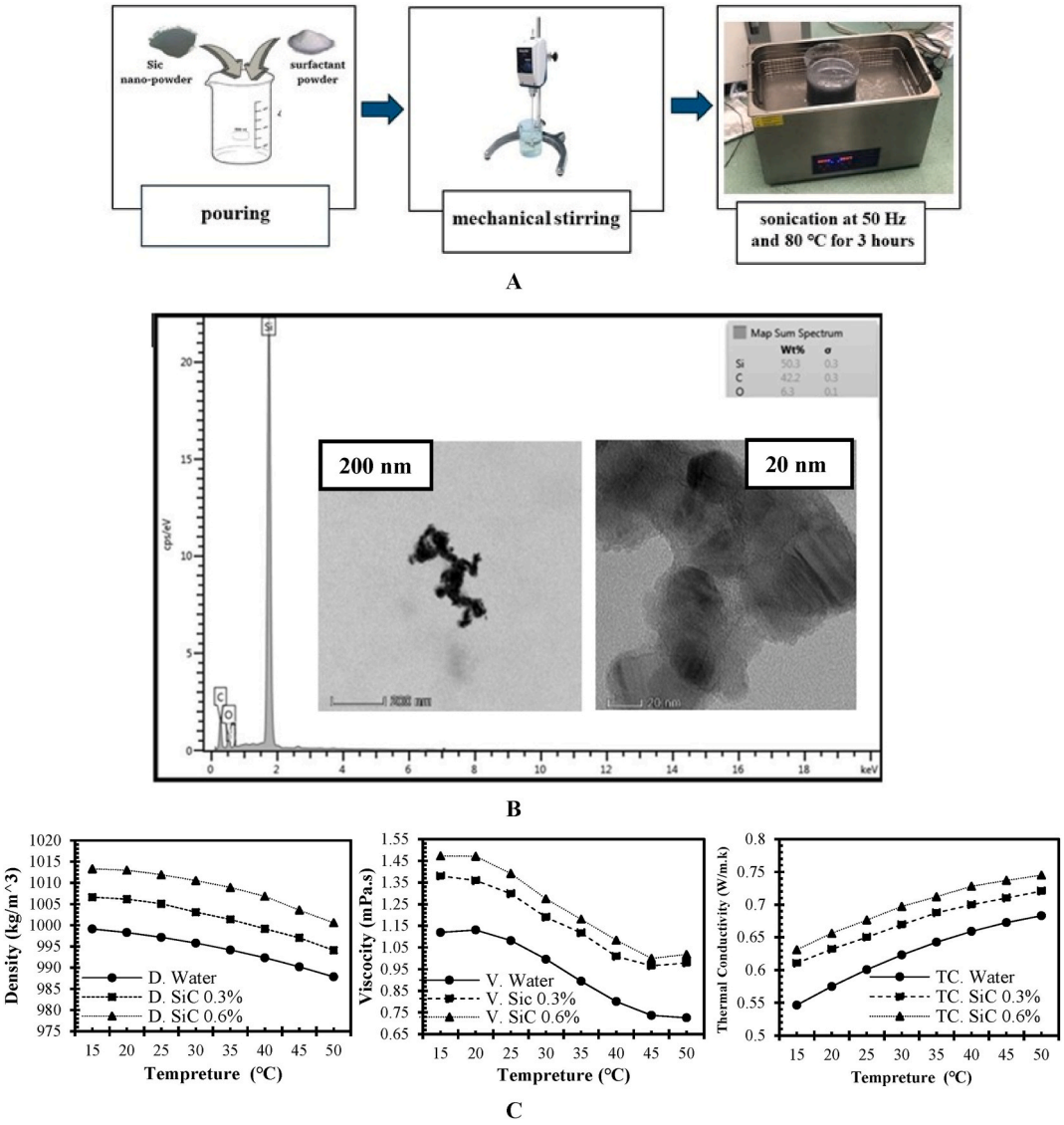



**Fig. 2.** (A) The preparation of the NF, (B) the TEM and FESEM images of the N, and (C) the thermophysical properties of the NF.

**Table 3**
The thermophysical properties of the PCM.

| PCM (paraffin wax) details | |
|---|---|
| Melting range | 45–49 °C |
| Density in liquid form | 836 (kg/m$^3$) |
| Density in solid form | 936 (kg/m$^3$) |
| Thermal conductivity | 0.24 (W/m°C) |
| Specific heat in solid form | 2100 (J/kg °C) |
| Specific heat in liquid form | 2100 (J/kg °C) |

K-type thermocouples recorded PV surface temperatures under solar irradiance of 400–1000 W/m$^2$. The MP-11 I-V checker tracked electrical outputs. Active components like the motor and chiller required external power, but pump energy consumption was subtracted for accuracy. Discharged fluid is not recycled in outdoor settings, so chiller power consumption is typically disregarded.

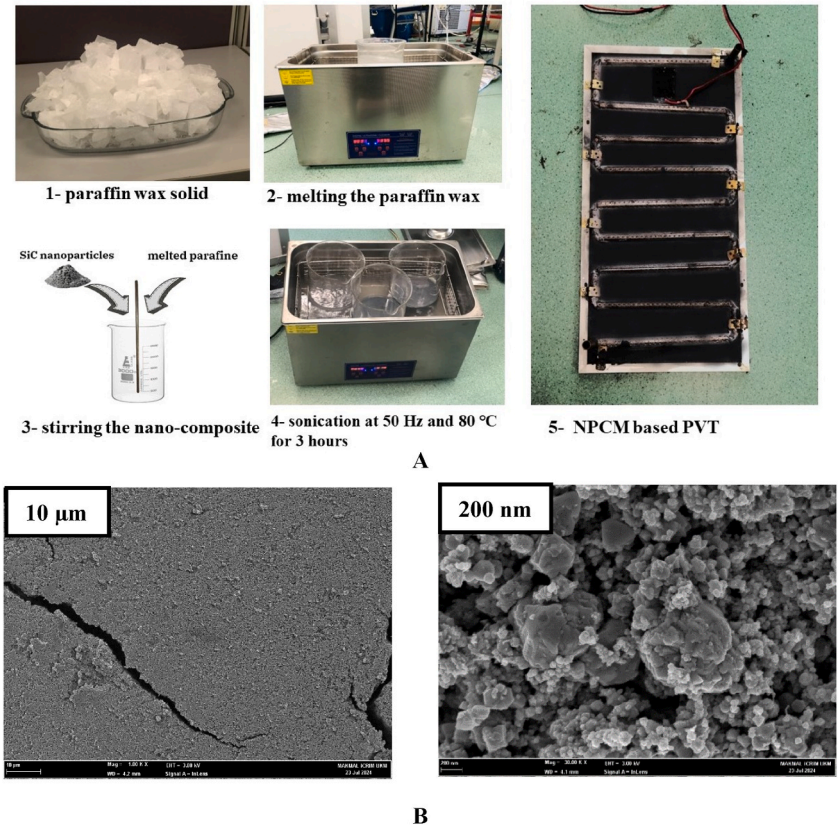



**Fig. 3.** (A) The preparation steps of the NPCM and (B) the FESEM images of the NPCM.

### *2.3. The mathematical models and methods used in the study*

#### *2.3.1. Heat transfer characterization models*

Eqs. (7) and (8) are used to calculate velocity and Reynolds number, which can be useful in future calculations [37,38]. In Eqs (7) and (8), $u$ is velocity, $\dot{m}$ is mass flow rates, ρ is density, A is area, Re is Reynolds Number, $D_h$ is hydraulic diameter, and $\mu$ is dynamic viscosity.

$$u = \frac{\dot{m}}{\rho A} \tag{7}$$

$$Re = \frac{\rho D_h u}{\mu} \tag{8}$$

After determining the Reynolds number, the heat gain can be calculated using Eq. (9). This heat gain is then used to compute the convective heat transfer coefficient by Eq. (10). With the convective heat transfer coefficient established, the Nusselt number can be readily calculated using Eq. (11). [37,38]. Parameters like $Q_u$ is useful heat gain, CP is specific heat capacity, T is temperature, h is convective heat transfer, q is again heat gain, and *Nu* is Nusselt number.

$$Q_u = \dot{m}\, CP\, (T_{out} - T_{in}) \tag{9}$$

$$h = \frac{q''}{T_{wall} - T_{ref}} \tag{10}$$

$$Nu_i = \frac{h_i D_h}{\lambda} \tag{11}$$

A comprehensive understanding of the tube's thermal performance requires more than just the Nusselt number calculation; it also involves conducting a pressure drop analysis and calculating the friction factor using Eq. (12) [37,38]. In the equation, ʃ is the friction factor, $\Delta P$ is pressure drop, L is the length of the channel.

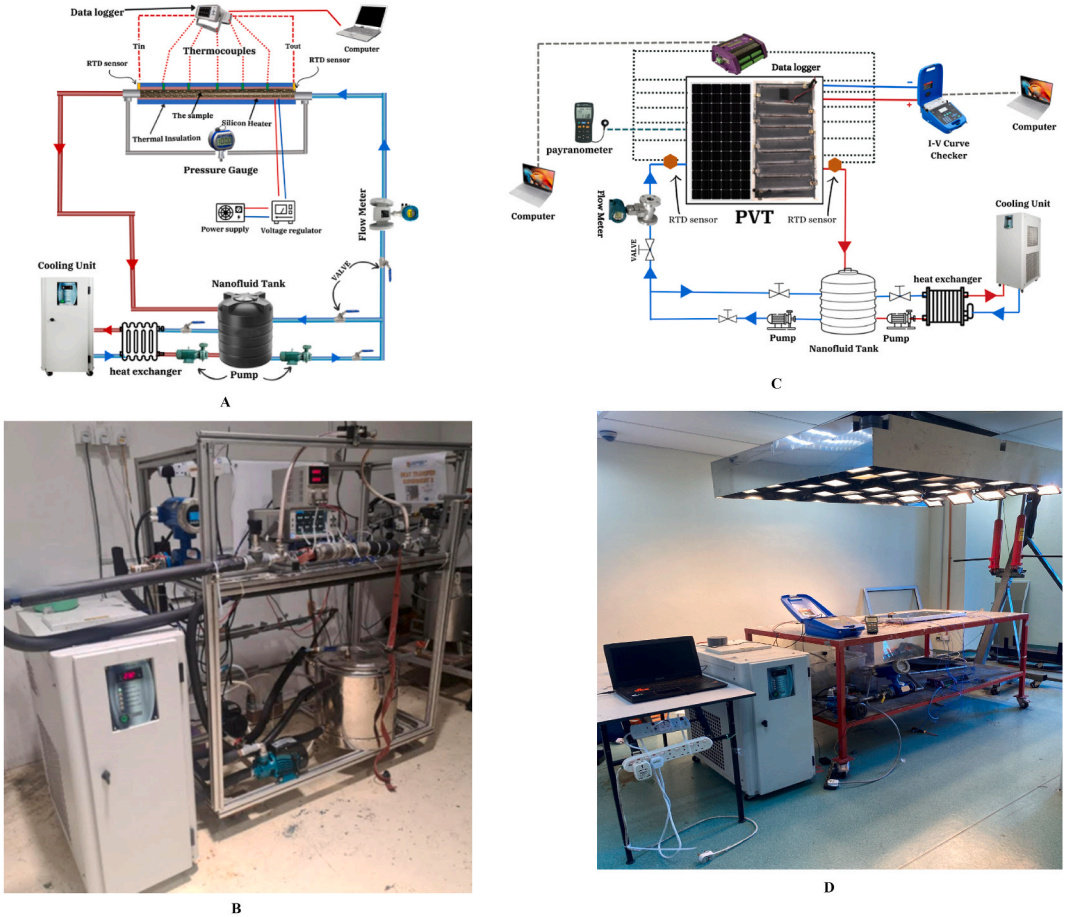



**Fig. 4.** The schematic diagram and experimental setup of the first experiment (A and B) the schematic diagram and experimental setup of the second experiment (C and D).

$$\int = \frac{2\Delta P}{\rho u_i^2}\frac{D_h}{L} \tag{12}$$

Finally, a balance must be struck between the Nusselt number and the Friction factor to determine which sample significantly improves thermal performance. THPF is a correlation that connects the Nusselt number and Friction factor, as presented in Eq. (13). THPF provides a valuable result that can guide the final decision regarding the tube's thermal performance [37,38].

$$\mathrm{THPF} = \frac{(\mathrm{Nu_i}/\mathrm{Nu_{i)}}}{(\int/\int)^{\frac{1}{3}}} \tag{13}$$

### 2.3.2. Energy models

The study initially utilized the energy balance equations outlined in Eq. (14)–(19) [39,40]. According to these equations, the total incoming energy, comprising solar energy and the energy of the incoming fluid, must equal the total outgoing energy. This outgoing energy includes the electrical energy output, the energy of the outgoing fluid, and any losses. In Eq (14)–(19), $A_{pv}$ is the area of the PV, $\tau_{gl}$ is transmissivity of the glass, $\alpha_{cell}$ is the absorbability of the PV cell, I is current, V is voltage, FF is the fill factor, $I_{sc}$ is current of short circuit, $V_{oc}$ is the voltage of the short circuit, and $\eta_{el}$ is electrical efficiency.

$$\sum \dot{E}_{in} = \sum \dot{E}_{out} + \sum \dot{E}_{loss} \tag{14}$$

$$\dot{E}_{sun} + \dot{E}_{mass\ in} = \dot{E}_{el} + \dot{E}_{mass\ out} + \dot{E}_{loss} \tag{15}$$

$$\dot{E}_{sun} = I * A_{pv} * \tau_{gl} * \alpha_{cell} \tag{16}$$

$$\dot{E}_{el} = I_{sc} * V_{oc} * FF \tag{17}$$

$$FF = \frac{I_{max} V_{max}}{I_{sc} * V_{oc}} \tag{18}$$

$$\eta_{el} = \frac{\dot{E}_{el}}{\dot{E}_{sun}} = \frac{I_{sc} * V_{oc} * FF}{I * A_{pv} * \tau_{gl} * \alpha_{cell}} \tag{19}$$

Eq. (20)–(22) calculate the thermal efficiency, while Eq. (16) indicates that the system's total energy efficiency is the sum of its thermal and electrical efficiencies [39,40].

$$\dot{E}_{mass\ out} - \dot{E}_{mass\ in} = \dot{E}_{th} = \dot{m} * C_p (T_{out} - T_{in}) \tag{20}$$

$$\eta_{th} = \frac{\dot{E}_{th}}{\dot{E}_{sun}} = \frac{= \dot{m} * C_p (T_{out} - T_{in})}{I * A_{pv} * \tau_{gl} * \alpha_{cell}} \tag{21}$$

$$\eta_t = \eta_{el} + \eta_{th} \tag{22}$$

It is important to note that the total energy output from the system is considered valid only after subtracting the energy required to operate the pumps. Eq. (23)–(25) calculate the energy losses associated with the pumps, providing an updated electrical energy efficiency [39,40]. In Eq (23)–(25), $\eta_p$ is the efficiency of the pump, and $\rho$ is the density of the pumped fluid.

$$\dot{E}_{el} = \dot{E}_{el} - \dot{E}_p \tag{23}$$

$$\dot{E}_p = \frac{\dot{m}\, \Delta P}{\rho \eta_p} \tag{24}$$

$$\eta_{el} = \frac{\dot{E}_{el} - \dot{E}_p}{\dot{E}_{sun}} \tag{25}$$

### 2.3.3. Sustainability analysis

*2.3.3.1. The exergy-based sustainability analysis.* Even though the previous section quantifies the total electrical and thermal energy the systems can produce, the literature prefers to conduct an exergy analysis to determine how much of the produced energy can be converted into useful work. Since electrical energy is a high-grade energy, almost all of it can be converted into useful work. Thus, electrical energy and exergy values in PVT systems are very close. Nevertheless, thermal energy is low-grade, meaning only a small portion can be converted into useful work. Recorded values of thermal exergy are usually very low. Even though this study has conducted a detailed exergy analysis, including total exergy ($\psi$) and total exergy destruction ($EX_{loss}$), its details will not be presented. Instead, this study will present the results in terms of sustainability index (SI) and improvement for potential (IP), metrics that not only tell how much of the produced energy is useful but also reveal how sustainable are the examined systems [41].

$$SI = \frac{1}{1 - \psi} \tag{26}$$

$$IP = (1 - \psi) EX_{loss} \tag{27}$$

*2.3.3.2. Economic analysis.* The economic feasibility of the studied systems will be handled by comparing the total amount of energy that the systems produce over their life cycles to the total amount of energy required overall their life cycle. Eq. (28)–(30) will calculate the annual energy production of the modules while the total energy required is calculated using Ecoinvent in SimaPro [42]. The study has accumulated a huge amount of data about all the components, such as pipes, PV modules, nanoparticles, paraffin wax, and others, with their respective details, such as size and weight. Also, processes such as welding, sonication, mixing, and others have been included. Components that are necessary for real-life applications, such as batteries and inverters, are also included. Moreover, energy required for material subtraction, transportation, and recycling are included as well. Creating such a database, the software has enabled a detailed calculation of the total amount of energy required, known as cumulative energy demand CED in the literature. In Eq (28)–(30), $G_t$ is solar irradiance, $dg_e$ is electrical degradation rate of the PV, and $dg_{th}$ is the degradation rate of the thermal collector.

$$En_{total} = E_{ele} + E_{th} \tag{28}$$

$$En_{ele} = \left(365 \times G_t \times \eta_{ele} \times A_{pv}\right)(1 - dg_e)^{t-1} \tag{29}$$

$$En_{th} = (365 \times G_t \times \eta_{th} \times A_{th})(1 - dg_{th})^{t-1} \tag{30}$$

Once AEP and CED are calculated, the deciding factor in determining the economic feasibility of the modules are cost of energy ($C_e$) and energy payback time (EPBT), which are calculated using Eq (31)–(38), where P is the initial value of the modules calculated to be $

20, 26, 28, and 30, S is the salvage values assumed to be $ 2, 2.4, 2.6, and 2.8, i is the interest rate assumed to be 8 %, and n is the lifespan claimed to be 25 years by the manufacturer. The remaining parameters can be defined as Ac is the annual cost, FAC is the first annual cost, AMC is the annual maintenance cost, ASV is the annual salvage value, CRF is the capital recovery factor, SFF is the sin fund factor, Ce is the cost of electricity, and EPBT is the energy payback time.

$$AC = FAC + AMC - ASV \tag{31}$$

$$FAC = CRF \times P \tag{32}$$

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \tag{33}$$

$$AMC = 0.15 \times FAC \tag{34}$$

$$ASV = S \times SFF \tag{35}$$

$$SFF = \frac{i}{(1+i)^n - 1} \tag{36}$$

$$C_e = \frac{AC}{AEP} \tag{37}$$

$$EPBT = \frac{CED}{AEP} \tag{38}$$

*2.3.3.3. Environmental impacts.* The study assesses the systems' environmental impact by comparing their $CO_2$ emission and mitigation capacity. The study uses Eq. (39) to calculate the total amount of $CO_2$ mitigated by each module while it uses the IPCC 2013 methodology in SimaPro software to assess the Global Warming Potential over a 100-year horizon (GWP 100a) [43]. This impact category measures the climate effects of different materials and processes by comparing their contributions to global warming relative to $CO_2$, allowing for a comparative evaluation of various greenhouse gases. The same data based described in the calculation of CED is used here. Once $CO_2$ emission and mitigation are calculated, the difference between them is a good indication of whether the systems are environmentally friendly or not.

$$\varphi_{co2} = \frac{En_{total} \times N \times \alpha_{co2}}{1000} \tag{39}$$

### 2.4. The uncertainty analysis

The study calculated the uncertainty of the two experiments using Table 4 and Eq. (34) [31]. It should be noted that the percentage of error recorded for the experiments are with the acceptable range which is usually reported to be below 3 %. Aside from the recorded uncertainty, the study did not have any other possible sources of errors.

$$W_R = \left[ \left( \frac{\partial R}{\partial X_1} W_1 \right)^2 + \left( \frac{\partial R}{\partial X_2} W_2 \right)^2 + \ldots \left( \frac{\partial R}{\partial X_n} W_n \right)^2 \right]^{0.5} \tag{34}$$

$$W_R = \left[ (0.2)^2 + (0.15)^2 + (0.15)^2 + (0.5)^2 + (0.08)^2 + (0.2)^2 \right]^{0.5}, W_R = 0.617\%$$

$$W_R = \left[ (0.01)^2 + (0.1)^2 + (1.5)^2 + (0.5)^2 + (0.01)^2 + (1)^2 \right]^{0.5}, W_R = 1.81\%$$

**Table 4**
The uncertainty of the devices used in the experiments.

| Devices | Recorded variables | Percentage of error |
|---|---|---|
| RTD probe | Temperature difference | ±0.01 |
| K-type thermocouple | Surface temperature | ±0.1 |
| TES solar meter | Solar irradiance | ±1.5 |
| DHYB-800 digital flow gauge | Mass flow rate | ±0.5 % |
| DT80 data logger | Cell temperature | ±0.01 |
| MP-11 I-V analyser | Electrical output | ±1.0 |
| Digital Differential Pressure Gauge | Pressure drops | 0.08 % |
| Cooling Unit | Water temperature | 0.2 % |

## 3. Results and discussion

### 3.1. The heat transfer characterization of the absorber tubes

The Nusselt number and Friction Factor of all the studied absorber tubes were calculated using the adopted methodology. As it is known, in the desired scenario, a sample will maximize the Nusselt number and minimize the Friction Factor. However, the studied samples of this study always recorded increased Nusselt number and Friction Factors. Thus, THPF was calculated to determine the best sample and quantify the enhancement percentage. Analyzing the ST, it was realized that increasing the Reynolds number causes an enhancement in the Nusselt number from 16.85 to 72.26, while the Friction Factor declined from 0.156 to 0.0485. The same transition led to Nusselt number enhancement from 22.013 to 85.710, corresponding to decreases from 0.161 to 0.056 in the Friction Factor in the DPAT case. Compared to the ST, there was enhancement of 29 %–12 % in THPF. When a coil was inserted in the DPAT and Reynolds number increased, Nusselt number values of 25.272–94.567, corresponding to declines from 0.173 to 0.062 in Friction Factor, were recorded. The enhancement in THPF was 45 %–20 % compared to the ST. Replacing the coil with a twisted tape, the Nusselt number increased from 29.868 to 113.480 with an increased Reynolds number, corresponding to the Friction Factor dropping from 0.186 to 0.067. The twisted tape obtained an enhancement of 67 %–40 % in THPF. Finally, inserting a coil and a twisted tape simultaneously led to an enhancement from 38.828 to 147.524 in the Nusselt number, while the Friction Factor values dropped from 0.205 to 0.074. The enhancement in THPF was 110 %–77 % for the coiled twisted tape. The main reason for the gradual enhancement recorded in the THPF of the samples is the incorporation of more passive cooling techniques which contribute to heat transfer through expanding area, generating more turbulence in flow, inducing vortex formation, etc. Therefore, it can be concluded that the DPATCTT records the best THPF values. Generally, the effect of the Reynolds number was negative on the THPF of all the samples. Fig. 5 summarizes all the results. The obtained results of this study can be compared to the results obtained by references number [22and31].

### 3.2. The effects of nanofluid on the heat transfer characterization of the absorber tube

Replacing water by 0.3 % VC of SiC-enhanced NF with the best sample led enhancements from 46.594 to 177.030 in the Nusselt number, corresponding to the Friction Factor decline from 0.219 to 0.079 as the Reynold number escalated. Such records led to an enhancement from 147 % to 108 % in THPF. Moreover, replacing 0.3 % VC of SiC-enhanced NF with 0.6 % VC of SiC-enhanced NF, the

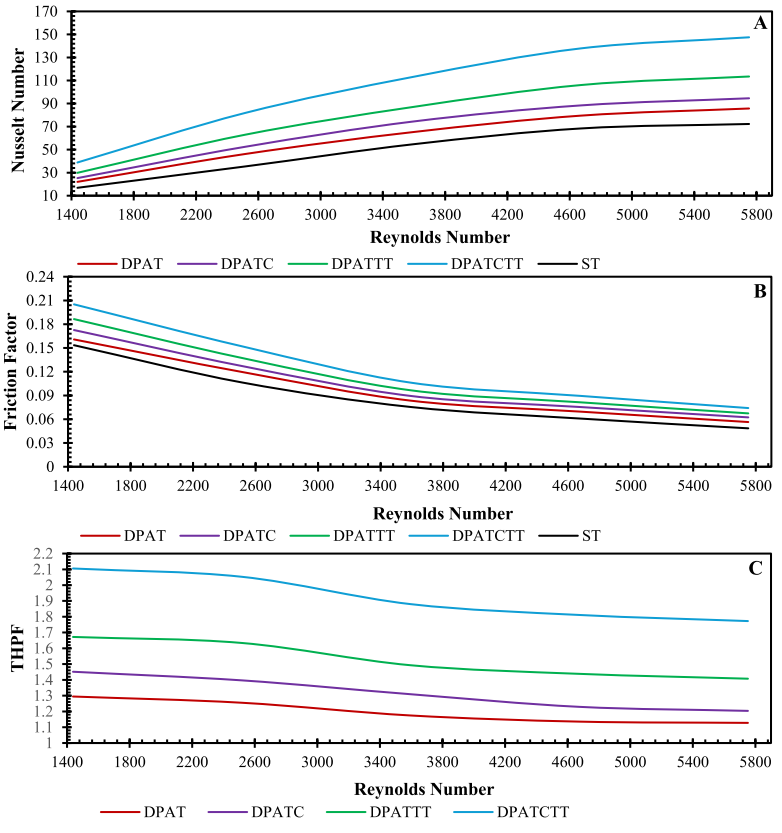



**Fig. 5.** The heat transfer characterization of the absorber tubes (A) *Nu*, (B) F, and (C) THPF.

system recorded Nusselt number from 58.243 to 221.286, while the Friction Factor decreased from 0.237 to 0.085 as Reynold number increased. The mentioned enhancements increased the THPF of the sample from 300 % to 253 %. Fig. 6 summarizes the results. The main reason for the recorded enhancements can be attributed to the fact that NFs have nanoparticles incorporated into their structure, which facilities the process of heat absorption significantly. The obtained results of this section can be validated by comparing the results with those obtained by reference number [22].

### 3.3. The surface temperature analysis of the photovoltaic thermal systems

The experiment was conducted in six distinct phases, each utilizing a different cooling medium: water, 0.3 % and 0.6 % VC of SiC-enhanced NFs, water with NPCM, and 0.3 % and 0.6 % VC of SiC-enhanced NFs with NPCM. Fig. 7 presents the surface temperatures recorded for each phase with different solar irradiance and mass flow rates. Flow rates significantly impact PV surface temperatures. For instance, under a 1000 W/m$^2$ solar irradiance, increasing flow rates from 0.01 to 0.085 kg/s using water leads to surface temperatures of 69.22 °C, 68.24 °C, 67.38 °C, 66.21 °C, 65.31 °C, and 64.26 °C, corresponding to an optimal surface temperature reduction of 7.2 %. A notable temperature reduction is observed when water and NFs are compared to the PV module without cooling. When cooled with water, 0.3 % VC of SiC-enhanced NF and 0.6 % VC of SiC-enhanced NF at a flow rate of 0.08 kg/s, temperatures drop to 64.26 °C, 57.06 °C, and 55.53 °C, corresponding to reductions of 34.3 %, 41.7 %, and 43.3 %, respectively. Incorporating NPCM with the cooling mediums achieves further reductions. Water with 1 % VC of SiC-based PCM, 0.3 % VC of SiC-enhanced NF with NPCM, and 0.6 % VC of SiC-enhanced NF with NPCM result in temperatures of 52.18 °C, 50.11 °C, and 48.83 °C, representing decreases of 46.7 %, 48.8 %, and 50.1, respectively.

### 3.4. The thermal energy efficiency analysis of the PVT system with various coolants

The study assessed the thermal energy efficiency of the system, highlighting its relationship with flow rate, solar irradiance, and types of coolant. Mass flow rates and various coolants demonstrated a positive impact on thermal efficiency, while the effect of solar irradiance is only positive till 1000 W/m$^2$, where it encounters a decrease. Employing water as a coolant at a solar irradiance of 400 W/m$^2$ and increasing the flow rate from 0.01 to 0.085 kg/s, improved thermal efficiency from 69.72 % to 74.77 % (a 6.8 % increase). This improvement is attributed to faster fluid circulation, which enhances heat exchange between the cooler liquid and warmer surfaces, effectively lowering surface temperatures. However, the impact of irradiance intensity on thermal efficiency is more complex. At a mass flow rate of 0.08 kg/s, thermal efficiencies using water reached 74.77 %, 76.61 %, 77.35 %, and 75.73 % under irradiances of 400–1000 W/m$^2$, respectively. The efficiency drop at 1000 W/m$^2$ stems from inadequate cooling to sustain optimal performance. At a solar irradiance of 800 W/m$^2$ and a mass flow rate of 0.085 kg/s, thermal efficiency showed a significant positive trend, rising from 77.35 % with water to 87.65 % (an 11.8 % increase) when transitioning to 0.3 % VC of SiC-enhanced NF, 0.6 % VC of SiC-enhanced NF,

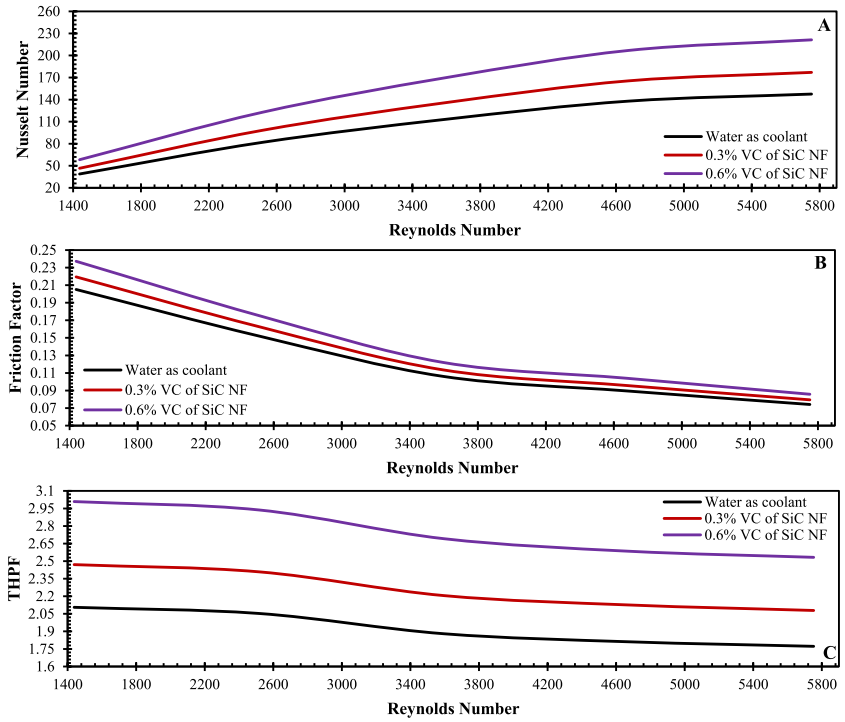



**Fig. 6.** (A) Nusselt number, (B) frication factor, and (C) THPF of the used NFs.

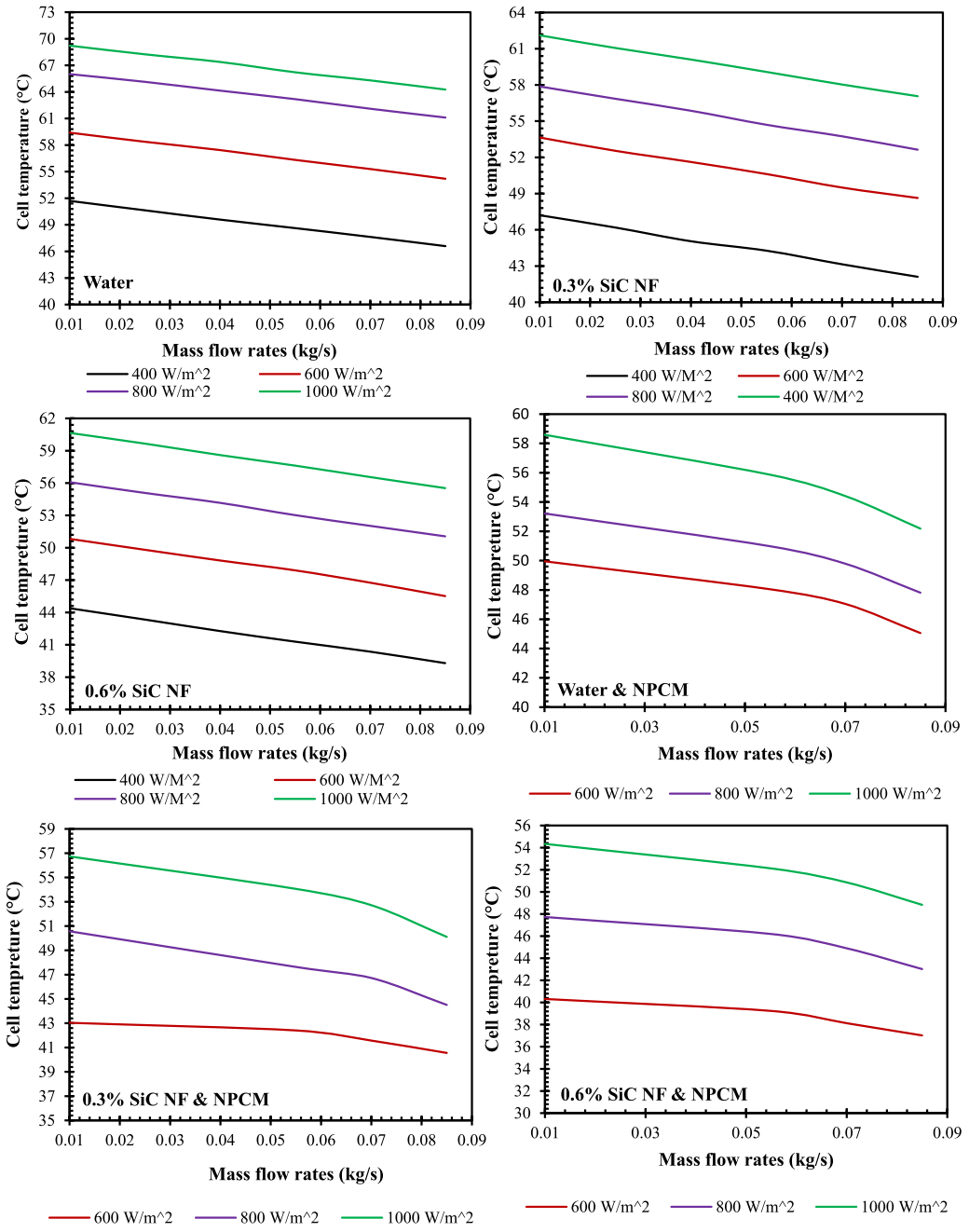



**Fig. 7.** The surface temperature analysis of the system employing varying primary conditions and coolant types.

and finally 0.6 % VC of SiC-enhanced NF combined with NPCM. Fig. 8 illustrates these findings. The highest thermal efficiency of 87.65 % (a 21.3 % improvement compared to the lowest case) was achieved using 0.6 % VC of SiC-enhanced NF with NPCM at a flow rate of 0.085 kg/s and a solar irradiance of 800 W/m$^2$. As was explored previously, the enhancements are all ascribed to enhanced surface area, more turbulence generation, inducing vortexes, and others created by the proposed design of this study.

### *3.5. The electrical energy efficiency analysis of the photovoltaic thermal systems*

The decline in surface temperature and increases in thermal efficiencies caused electrical efficiency to increase significantly. It is revealed that there is a positive correlation between electrical efficiency and mass flow rates. As flow rates increase over the

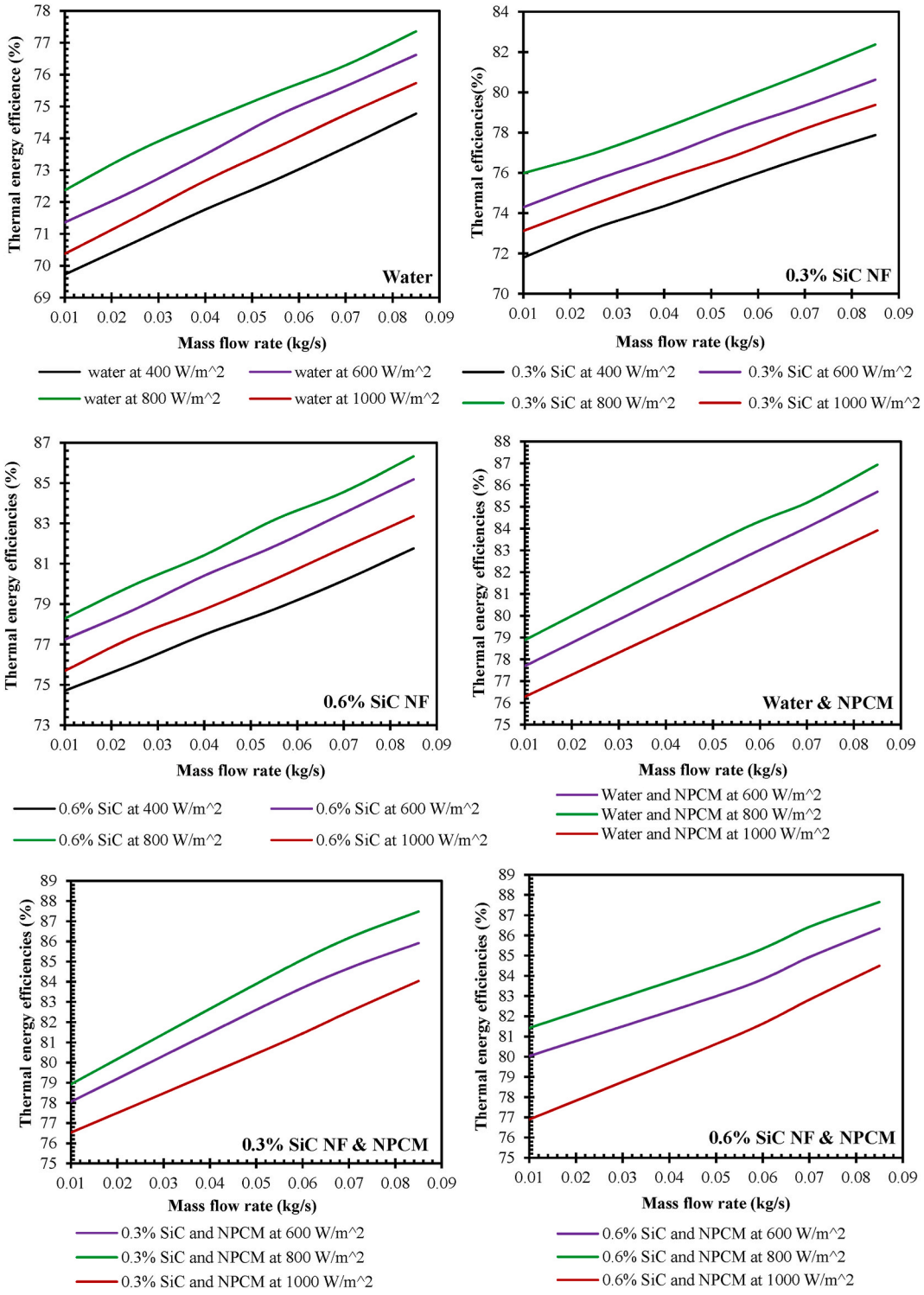



**Fig. 8.** The thermal energy efficiency analysis of the systems employing varying primary conditions and coolant types.

determined range of the study using water and 400 W/m$^2$, the electrical efficiency increases from 8.271 % to 8.326 % (0.6 % increase), ascribed to the effect where increased circulation accelerates the exchange between the heated surface and the cooler fluid. Nevertheless, the correlation between electrical efficiency and irradiances needs more examination. As solar irradiance increases from 400 to 800 W/m$^2$ at a mass flow rate of 0.08 kg/s and employing 0.6 % VO of SiC-enhanced NF, electrical efficiencies increase from 9.828 %,

9.975 %, and 11.099 %. Nevertheless, increasing the solar irradiance to 1000 W/m$^2$ will have a negative impact, reducing electrical efficiency to 10.670 %. This decrease can be traced back to the fact that as solar irradiance gets as high as 1000 W/m$^2$, the cooling mechanism of the system declines to maintain the system working at its ideal condition. However, transitioning from bare PV to water, 0.3 % VC of SiC-enhanced NF, 0.6 % VC of SiC-enhanced NF, and 0.6 % VC of SiC-enhanced NF combined with NPCM improves electrical efficiency to 10.628 %, 11.044 %, 11.099 %, and 11.688 %, corresponding to increases of 25.9 %, 28.7 %, 29 %, and 32 %, respectively. Fig. 9 summarizes the results.

### 3.6. The sustainability analysis of the photovoltaic thermal systems

As discussed in the methodology section, the study calculates the Sustainability Index SI and Improvement Potential IP to make sure

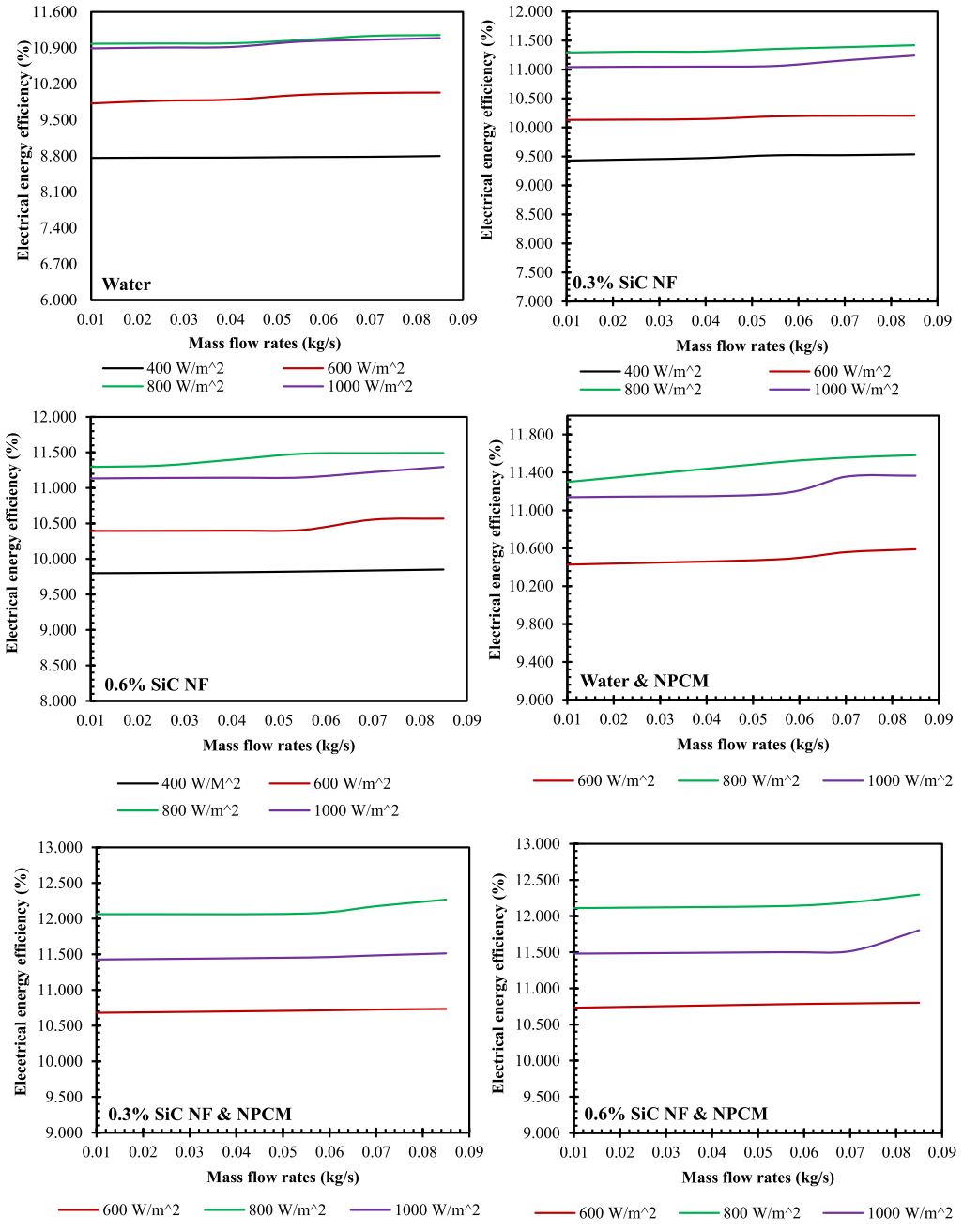



**Fig. 9.** The electrical energy efficiency analysis of the systems employing varying primary conditions and coolant types.

whether transitioning from PV alone to PVT with NF and NPCM was a good choice regarding becoming more sustainable and converting more energy into useful work. The SI shows values such as 1.15, 1.16, and 1.19, meaning that the transition indeed makes the systems more aligned with the principles of sustainability. Moreover, the IP indicates values such as 92.50, 88.52, and 86.85 $W/m^2$, demonstrating that the transition strengthens the systems by reducing waste and converting more energy into useful work.

Once the differences between total energy output and input showed great differences, as is the case with the studied system of this research, it is a good indication that the system will be cost-effective, depending on factors such as the initial cost of products, interests rate of countries, maintenance cost, and salvage values. Nevertheless, Calculating total energy differences between output and input to be 1.55, 1.78, and 2.11 MWh is a good sign that the systems are cost-effective regardless of the effect of the factors mentioned. Considering the mentioned factors, the study recorded energy cost improvement of 8.63 %, 9.685, and 14.09 % while the energy payback enhancements are 12.5 %, 12.98 %, and 17.44 %, respectively, showing great enhancement in cost of energy and the time required to pay back the total energy needed. It should be noted that the CED performed in this study is very comprehensive (as shown in the methodology section) including all factors contributing to the total energy required in the design of modules compared to other studies of literature in which only assumptions are made about CED. Therefore, the long payback period of 14–17 years is normal as compared to lower values reported in the literature.

Even though environmental assessment of any product or process can be detailed, literature usually regards PVT systems, which are better at mitigating $CO_2$ than the total amount of $CO_2$ they emit, as being environmentally feasible. The difference between $CO_2$ mitigation and $CO_2$ emissions in the cases of this study are 0.48, 0.53, and 0.63 tons. Such great differences are good signs that the studied systems are environmentally friendly. while Table 5 summarizes the result of this section, Table 6 compares this study's results to other currently published articles in the field.

## 4. Conclusions

- Compared to an ST, the DPATCTT has the best thermal performance by recording the highest Nusselt number of 147.524, the Friction Factor of 0.074, and the THPF of 110 %. The cause is attributed to the usage of several passive cooling techniques, which expand contact area, vortex formation, and turbulence generation.
- Replacing water by 0.3 % VC of SiC and 0.6 % VC of SiC and employing DPATCTT increased the values of THPF from 147 % to 300 %, respectively. The reason for the obtained result is that the incorporation of nanoparticles facilitates the heat transfer process with a minimal amount of frictional loss.
- Generally, increasing mass flow rates increase the thermal and electrical efficiency of the PVT systems due to the fact that a faster flow rate brings cold water into contact with hot surface faster, while the effect of solar irradiances is positive till 1000 $W/M^2$, where it shows a decline in electrical and thermal energy efficiencies due to optimal cooling capacity being reached.
- The system attained optimal electrical and thermal efficiencies of 11.69 % and 87.65 %, respectively, under solar irradiance of 800 $W/m^2$, a flow rate of 0.08 kg/s, and with 0.6 % VC of SiC-enhanced NF combined with NPCM. These results represent improvements of 32 % and 21.2 %, respectively.
- The systems recorded sustainability indexes of 1.15, 1.16, and 1.19, while improvement potentials were 92.95, 88.53, and 86.85 W/m2, respectively. The obtained result can be attributed to the fact that the system maximizes total exergy production and minimizes exergy destruction as the effect of passive cooling techniques becomes more dominant.
- The systems also recorded total input energy of 2.65, 2.79, 2.82, and 2.91, while the total output energy was 0.724, 4.34, 4.58, and 5.02, respectively. As presented, the system's output energy production exceeds the system's input energy, which indicates that the proposed design will be cost-effective and implementable.
- The systems were accountable for omitting 0.56, 0.60, 0.61, and 0.62 tons of $CO_2$ while they could mitigate 0.18, 1.08, 1.14, and 1.25 tons of $CO_2$. Such data confirms that the proposed system is in alliance with sustainability principles adopted by governments and international organizations worldwide, making its implementation easier.

Aside from all the insightful details provided in this study, there are several limitations that future research should cover.

- Outdoor experiments should be conducted to reveal the effects of the new design under real-world conditions.
- A detailed economic and environmental analysis of the systems should be conducted to reveal their feasibility.
- Numerical studies should be conducted to maximize the effect of geometrical parameters.

## CRediT authorship contribution statement

**Hariam Luqman Azeez:** Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Adnan Ibrahim:** Visualization, Validation, Supervision, Methodology, Conceptualization. **Banw Omer Ahmed:** Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Sharul Sham Dol:** Writing – review & editing, Supervision, Methodology. **Ali H.A. Al-Waeli:** Visualization, Validation, Supervision, Methodology, Conceptualization. **Mahmoud Jaber:** Validation, Methodology, Data curation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

**Table 5**
The sustainability analysis of the PVT systems.

| Studied parameters | PV alone | PVT + new design + water | PVT + new design + NF | PVT + new design + NF + NPCM |
|---|---|---|---|---|
| Sustainability index | – | 1.15 | 1.16 | 1.19 |
| Improvement potential ($W/m^2$) | – | 92.50 | 88.53 | 86.85 |
| Input energy (MWH) | 2.65 | 2.79 | 2.82 | 2.91 |
| Output energy (MWh) | 0.724 | 4.34 | 4.58 | 5.02 |
| Cost of electricity ($/KWh) | 0.074 | 0.068 | 0.067 | 0.063 |
| Energy payback time (years) | 17.6 | 15.44 | 15.32 | 14.52 |
| $CO_2$ emissions (tons) | 0.56 | 0.60 | 0.61 | 0.62 |
| $CO_2$ mitigation (tons) | 0.18 | 1.08 | 1.14 | 1.25 |

**Table 6**
Comparing the results of some studies in the literature to the results of this study.

| Ref and years | Coolant types | Overall energy efficiency (%) |
|---|---|---|
| [44] & 2023 | Copper oxide-enhanced NF and PCM. | 76.3 |
| [45] & 2020 | A novel finned multi-block PCM and NF | 78.2 |
| [33] & 2023 | Twisted tube, NF, and NPCM | 94.3 |
| [34] & 2023 | Micro-fin, twisted tape, NF, and NPCM | 94.1 |
| [31] & 2024 | Ribs, coiled twisted tape, NF, and NPCM | 97.4 |
| [46] & 2025 | Square spring, NPCM, and NF | 92.6 |
| The current study | Dimples, petal arrays, coiled twisted tapes, NF, and NPCM | 98.2 |

influence the work reported in this paper.

## Nomenclature

| | | | |
|---|---|---|---|
| A | Area ($m^2$) | NPCM | Nanophase changing materials |
| AC | Annual cost ($) | N | Number of years |
| AMC | Annual maintenance cost | ND | New design |
| ASV | Annual salvage value | $\Delta P$ | Pressure drops |
| $Adg_e$ | Annual Model Degradation (%) | P | Initial price |
| CP | Specific heat (J/Kg K) | PV | Photovoltaics |
| CRF | Capitale recovery factor | PVT | Photovoltaic thermal |
| $D_h$ | Hydraulic diameter (m) | $Q_u$ | Heat obtained (W) |
| $\dot{E}_{out}$ | Energy out (W) | $Re$ | Reynolds Number |
| $\dot{E}_{loss}$ | Energy loss (W) | SD | Smooth tube |
| $\dot{E}_{mass-in}$ | Energy due to inlet fluid | SFF | Sink fund factor |
| $\dot{E}_{mass-out}$ | Energy due to outlet fluid | SI | Sustainability index |
| $\dot{E}_{sun}$ | Sun energy (W) | $T_{out}$ | Outlet temperature (°C) |
| $\dot{E}_p$ | Energy from pump($W/m^2$) | $T_{in}$ | Outlet temperature (°C) |
| $\dot{E}_{el}$ | Electrical energy | $T_a$ | Ambient temperature (°C) |
| $\dot{E}_{th}$ | Thermal energy | $THPF$ | Thermal hydraulic performance factor |
| $En_{ele}$ | Annual electrical energy production (kWh) | $V_{oc}$ | The voltage of open circuit (V) |
| $En_{th}$ | Annual thermal energy production (kWh) | $V_{max}$ | Max voltage (V) |
| $En_{total}$ | Annual total energy production (kWh) | **Greek Symbols** | |
| $EX_{loss}$ | Total exergy destruction ($W/m^2$) | ∅ | Volume fraction % |
| EPBT | Energy payback time (years) | $\rho_{nf}$ | Density of nanofluid ($kg/m^3$) |
| FF | Fill factor (%) | $\rho_{bf}$ | Density of based fluid ($kg/m^3$) |
| FAC | First annual cost | $\rho_{np}$ | density of nanoparticles ($kg/m^3$) |
| ∫ | Friction factor | $\mu_{nf}$ | Viscosity of the nanofluid (Pa. s) |
| $G_t$ | Annual solar irradiance (kWh/m2/day) | $\mu_{bf}$ | Viscosity of the based fluid (Pa. s) |
| $h_i$ | Convective heat transfer coefficient ($W/m^2K$) | $\eta_e$ | Electrical efficiency (%) |
| $I_{sc}$ | The short circuit current (A) | $\eta_{th}$ | Thermal efficiency (%) |
| I | Irradiance ($W/m^2$) | $\eta_t$ | Total efficiency (%) |
| i | Interest rate | $\tau_{gl}$ | transmissivity of the glass |
| $I_{max}$ | Max current (A) | ρ | Density ($Kg/m^3$) |
| IP | Improvement potential (W) | $\eta_p$ | Pump efficiency (%) |
| K | Thermal conductivity (W/m. K) | $u$ | Velocity (m/s) |
| $K_{np}$ | Thermal conductivity of nanoparticle (W/m. K) | $\alpha_{cell}$ | Cell absorptivity |
| $K_{bf}$ | Thermal conductivity of the based fluid (W/m. K) | $\Delta P$ | Pressure drops (pascal) |
| $K_{nf}$ | Thermal conductivity of the nanofluid (W/m. K) | $\psi$ | Total exergy |
| $Nu_i$ | Nusselt number | $\varphi_{co2}$ | Environmental parameter (tons) |
| N | Avogadro number | | |

## Data availability

Data will be made available on request.

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